FortyGuard Temperature Intelligence Layer for QGIS

Internal testing fix report: every issue raised by the FortyGuard team, its root cause, the fix shipped, and how it was verified (plugin versions 0.5.0 to 0.5.3)

Task	Update the QGIS integration issues reported in the internal testing sheet ("FortyGuard QGIS Plugin - Internal Testing")
Testing rounds	19 October 2026 (9 testers, macOS / Windows / Ubuntu) and 20 October 2026 (3 testers, Windows), plus an independent code review on 20 October 2026
Issues handled	49 rows: 26 from round one, 11 from round two, 12 from the code review; all fixed, none deferred
Versions shipped	0.5.0, 0.5.1, 0.5.2, 0.5.3 (GitHub releases with installable ZIPs); 0.5.3 is the version to publish on plugins.qgis.org
Verification	96 pure-Python and 105 QGIS-runtime automated tests (0 failed); Bandit and plugins.qgis.org validator clean; three real production requests re-checked after the fixes
Source	https://github.com/FortyGuard-Tech/qgis-fortyguard-heatmap (GPL-2.0-or-later)
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1. Summary

Nine colleagues tested plugin 0.4.2 from the official QGIS Plugin Repository against the production API on 19 October 2026 and recorded 34 rows in the internal testing sheet; three more colleagues added 12 rows on 20 October (Windows). The installation, Demo Mode, the direct comparison of all four analyses with the raw API (0 differences) and the Windows load/unload cycle passed. Every reported failure was traced to a root cause, fixed, covered by an automated test and released. An independent code review was then run on the whole plugin before the next round and found twelve further defects, which were fixed in 0.5.3.
One finding dominates: the plugin trusted the API documentation in places where the production API behaves differently. The documentation says data is available through 12 hours past the current time; in practice the most recent five to seven days are rejected. The documentation describes plan area caps; the API accepts any area, charges a flat 4,220 credits, and never finishes areas above a few hundred square miles. Completed jobs with zero tiles (areas outside the United States or smaller than one tile) are charged. The credits field the plugin displayed stays frozen after a billing cycle ends. The fixes therefore add client-side pre-checks, safer defaults and honest messages, and record these observations in the API contract document for the API team.

Highlights of what changed for users

- Dates default to one week ago; a request for recent days is explained in plain words and marked as free of charge.
- Before any paid request the plugin checks coverage (United States), area size, recent dates, duplicates and unfinished earlier requests, and states the expected cost for the user's plan.
- "Check key and credits" shows the live balance; it moved by exactly 4,220 credits per request in verification.
- Legends are readable: labels get as many decimals as the data needs, uniform results show one class, tiles without a value no longer distort the classes, peak-time hours fall in the right class, and the smooth surface is named and bounded.
- Paid results are never lost: Stop during submission keeps the activity ID, cut-off answers are retried, results are saved before styling, and any earlier activity ID can be fetched again with Check Again.
- Dialog polish: readable steps, no orphan labels, remembered analysis settings, narrow-window wrapping, a demo-specific stop message, in-place status updates, a plain description, and the development environment hidden in the public build.
- The plugin is no longer marked experimental: it has now run on macOS, Windows 11 and Ubuntu.

2. Testing rounds and scope

Round	Date	Testers	Platforms	Rows	Outcome
1	19 Oct 2026	Fares, Rohit, Arpit, Inji, Sarang, Aashan, Akhilesh, Fawad (8) + Mustafa	macOS 26/27 (QGIS 3.44.15), Windows 11 (3.44), Ubuntu 26 (3.40.15)	34	4 passed, 26 failures / observations, 1 blocked (static review) -> fixed in 0.5.0
2	20 Oct 2026	Shreya, Mudethir, Snehil	Windows 11 (QGIS 3.44.15 / 3.44)	12	1 passed, 10 failures / requests, 1 blocked (time-zone text to confirm) -> fixed in 0.5.2
Review	20 Oct 2026	Independent code review (fresh-eyes pass, headless dialog driving)	macOS (QGIS 3.44.15)	12	12 verified defects -> fixed in 0.5.3

What passed without change: installation and reopening on all three platforms; Demo Mode on all analyses; the direct API comparison of Snapshot, Exceedance, Persistence and Peak temperature time (payloads, tile IDs, geometries, values, statistics and saved files identical, 0 differences); "Generate" correctly ignored while a job is running; load, unload and reopen on Windows; encrypted key storage.

3. Root causes observed on the production API

These observations were made by the testers and confirmed by the fixes' own verification. They are not in the public documentation and are now recorded in docs/API_CONTRACT.md (section "Observed behaviour") for the API team.

Topic	Observed	Plugin response
Data latency	On 19 October the days 15-18 October were rejected with HTTP 400 "Temperature data is unavailable for one or more hours"; 14 October completed. Rejections were free.	Default date = today minus 7; warning for later windows; Calendar month clipped; plain-language message.
Range of days	The end date is inclusive; 1 July to 1 August (32 days) was accepted, ran seven minutes and failed without a reason.	Local rule: 31 days counting both dates; hint to use Calendar month.
Credits	Every completed request cost exactly 4,220 credits on Premium and Hackathon keys, for any area or resolution.	Observed cost shown for plans without a documented figure.
Empty results	Paris, London, open ocean, a 64 m2 polygon and a 10 m polygon all completed with 0 tiles and were charged.	Coverage outline check (block), tile-size warning, "empty" result kind that states the charge, local record file.
Area caps	Not enforced: 400 mi2 completed (103,099 tiles); 1,300 and 11,472 mi2 were accepted, charged, and the status endpoint answered HTTP 504 forever.	Warn above the plan allowance; block above 500 mi2; explain repeated 5xx.
Usage endpoint	cycle_remaining_credits stayed frozen across nine billed requests on a key whose cycle had ended; total_remaining_credits moved by 4,220 each time.	Read total_remaining_credits.
Idempotency	An identical request re-run after completion created a new activity and was charged again.	Request fingerprint stored in the sidecar; warning when it was already completed.
Gateway errors	Occasional HTTP 504 on submission for small areas; the same request succeeded on the next try.	Explained as server-side and uncertain, not as a key problem.

4. Issues, root causes and fixes

4.1 Round one (fixed in 0.5.0, released 20 October 2026)

#	Reported by	Issue	Root cause	Fix
1	Fares, Arpit, Inji, Aashan	Default date (yesterday) and recent dates fail with HTTP 400; message unexplained	Plugin trusted the documented "now + 12 h" limit	Default today - 7; latency warning; no_data kind with a plain explanation and "nothing was charged"
2	Aashan	Calendar month for the current month always fails	Clipped to now + 12 h	Clipped to the available days; explained
3	Fares	Range of days defaults confusing	start = end = yesterday	Valid 7-day default window
4	Aashan	32 inclusive days accepted; 7-minute server failure	Day count ignored the inclusive end date	Count both dates; hint to Calendar month
5	Aashan, Mudethir	Credits shown never move	Frozen cycle_remaining_credits read first	Live total_remaining_credits; low-credit warning on the same figure
6	Aashan, Fawad, Rohit	Paris / London / ocean / 64 m2 polygons: 0 tiles, charged, reported as "malformed"	No coverage or size check; empty result treated as a parsing error	Coverage outline (block / warn); tile-size warning; "empty" kind stating the charge; record file
7	Arpit, Fawad	Huge polygons accepted, charged, HTTP 504 forever	No area check; generic give-up message	Block above 500 mi2; warn above the plan allowance; explained message
8	Aashan	Identical request charged again	Only "Generate while active" was guarded	Fingerprint in the sidecar; warning in this session and from the output folder
9	Fawad	Stop during submit may lose the paid job's ID	The POST was aborted	Answer awaited, ID recorded, then stop
10	Fawad	New Generate after a timeout orphans the earlier job	Silent disposal	Unfinished activity pointed out in the confirmation
11	Fawad	Status body cut off mid-transfer stops polling	Fatal "malformed"; transport errors with a status were parsed	Retryable; network-layer errors retried
12	Akhilesh	Old activity 404 shown as "just submitted", 8 retries	Same policy for submit and resume	3 checks on resume, specific message
13	Aashan	"Message: Failed" repeats the status	Appended whenever present	Only when informative
14	Fawad	Transient HTTP 500 on submit; user not told it was free	Generic message	States no job was created and nothing was charged
15	Aashan, Fawad	Legend labels collapse ("21.9 - 21.9"); uniform results show seven identical classes	One-decimal labels; quantile on few distinct values	Decimals follow the spread; zero-width classes dropped; one class for uniform results
16	Fawad	Peak-time hours 4, 8, 12, 16, 20 in the previous class	QGIS matches both class bounds inclusively	Upper bound just below the next bound
17	Aashan	Surface legend "Band 1 (Gray)"; 0.21 C stretched over the whole ramp	No band description; min-max stretch	Named band; minimum colour span 1 C / 1 h; spread shown
18	Fawad	NaN written as literal NaN (invalid JSON)	json.dump default	Non-finite values become null
19	Fawad	Long summary: OSError; sidecar overwritten	Unbounded name; only .geojson checked	Capped stem; all sibling files checked
20	Sarang	Orphan "Start date" label in Calendar month mode	Label never hidden	Label toggled with the field

#	Reported by	Issue	Root cause	Fix
21	Sarang	Demo stop mentions a server job and a disabled Check Again	One handler for demo and live	Demo-specific message
22	Sarang	Getting-started steps run together	No line breaks	One step per line
23	Aashan	Analysis settings not restored after restart	Only presentation settings saved	Analysis, time option, threshold, resolution, times remembered
24	Akhilesh	Texts and buttons cropped at narrow widths	Fixed-width rows	Wrapping rows and labels; own row for the credits button
25	Akhilesh	Hackathon plan: no cost in the confirmation	Only documented plans quoted	Cost for the checked plan, observed cost otherwise
26	Fares, Shreya	Install needed "Show also experimental plugins"	experimental = True	experimental = False after Windows and Linux tests

0.5.1 added a "Legend classes" option (3 to 20, default 10) after the report review asked for more classes.

4.2 Round two (fixed in 0.5.2)

#	Reported by	Issue	Root cause	Fix
27	Mudethir	Lowest legend class "0.0 - 26.3 C" when one tile has no value	QGIS updateClasses() counts NULL as 0 (reproduced on macOS)	Breaks computed from the numeric values only
28	Mudethir	Default folder ignores OneDrive-redirected Documents	Home folder + "Documents"	Documents folder reported by the system
29	Shreya	Cannot check a stopped request after sending another	Only the last job could be polled	Editable Activity ID field; Check Again on any pasted ID; IDs logged
30	Shreya	HTTP 504 on submit; same request worked next time; "reload the key?"	Message did not say it was server-side	Gateway errors explained: server-side, not the key, outcome uncertain
31	Snehil	"Processing" repeated every poll	One line per check	Status line updates in place; log only on change
32	Snehil	Appearance option refers to "the dashboard"	Wording	"Smooth surface (interpolated estimate) + exact tiles layer hidden"
33	Snehil	Developer settings expose the development environment	Selector always built	Hidden unless FORTYGUARD_DEV_ENVIRONMENT=1; "Advanced settings"
34	Snehil	Plugin description cluttered	Long paragraphs	Short sections with bullets
35	Snehil	Time-handling helper text to be confirmed	-	Unchanged: matches the API team's confirmation of 13 October; open for re-confirmation
36	self-review	"18.51 - 18.51 C" labels with 10 classes on a 0.27 C spread	Fixed decimals per spread	Decimals increase until every label is distinct

4.3 Independent code review (fixed in 0.5.3)

#	Severity	Defect	Fix
38	High	Check Again on an earlier activity used the latest request's analysis type, threshold and area; a stopped Exceedance fetched after a Temperature run errored and the paid GeoJSON was not saved	Per-activity request context; the API's stats_data.analytic_type is authoritative
39	High	A cancelled confirmation or blocked pre-check overwrote the previous job's context	Context applied only when the job starts
40	Medium	Restyle after the layer was removed raised a Python error	Layer resolved through the project; clean message

#	Severity	Defect	Fix
41	Medium	Restyle changed the hidden tiles but not the visible surface	Surface rebuilt for the new measure
42	Medium	Schema check raised before saving; message claimed the file was saved	Save first, then style; path in the message
43	Medium-low	Zero-width first class forced four-decimal labels	Zero-width classes merged into neighbours
44	Low-medium	PolygonZ/M areas sent 3-D coordinates	Z/M dropped
45	Low	Check Again after a demo run replayed the fixture as if live	Demo IDs refused
46	Low	Same-job Check Again used the key from submission time	Client rebuilt when the key differs
47	Low	Date pickers allowed tomorrow; the payload rejected it before noon	Maximum date = today
48	Low	OSError while saving lost the paid result with a traceback	Caught; message with the activity ID
49	Low	Duplicate warning pointed at an empty-result record as a loadable file	Separate wording
-	Hardening	Enter could trigger Generate; a button re-enabled mid-run; unfinished activity cleared too eagerly; surface memory on large areas	Non-default buttons; busy check; per-activity clearing; adaptive pixel grid

5. Verification

5.1 Automated

Check	Result
scripts/run_tests.sh (0.5.3)	96 pure-Python tests and 105 QGIS-runtime tests, 0 failed (1 keychain test skipped headless). New suites: tests/test_preflight.py and DialogLiveFlowTests, which drives the real dialog against a scripted local HTTP server that plays the FortyGuard API (Stop during submit, cut-off bodies, 404 on resume, Check Again across requests, cancelled confirmations, restyle after layer removal).
Security scan	Bandit clean (the same scan plugins.qgis.org runs); scripts/package.sh refuses to build otherwise.
Repository validator	scripts/validate_package.py: PASS for fortyguard_heatmap-0.5.3.zip (metadata, URLs, structure).
Dialog at narrow widths	Rendered offscreen at 560 px and 680 px: all rows wrap, nothing cropped.

5.2 On the production API (20 October 2026, Hackathon key shared with the team)

Check	Result
Request for yesterday (free)	HTTP 400 classified as no_data: "no temperature data yet ... choose dates ending on or before 2026-10-13 ... Nothing was charged".
1 July to 1 August	Blocked locally before submission with the day count and the Calendar month hint.
Full dialog run with the default date (today - 7), twice	Both completed (activities b291e768, c2a98efa): 150 tiles, layers added, files saved without the key, project reopened. The confirmation showed the plan-specific cost and "inside documented coverage".
Credits display	8,538,680 before the second run, 8,534,460 after: exactly 4,220, while the frozen cycle figure stayed at 9,341,680 (the value the testers saw).
San Jose, 4 km square, 60 m, 15-21 July 2024	Completed: 4,290 tiles; used for the review screenshots (section 6).
Cost of the verification	3 completed requests x 4,220 = 12,660 credits.

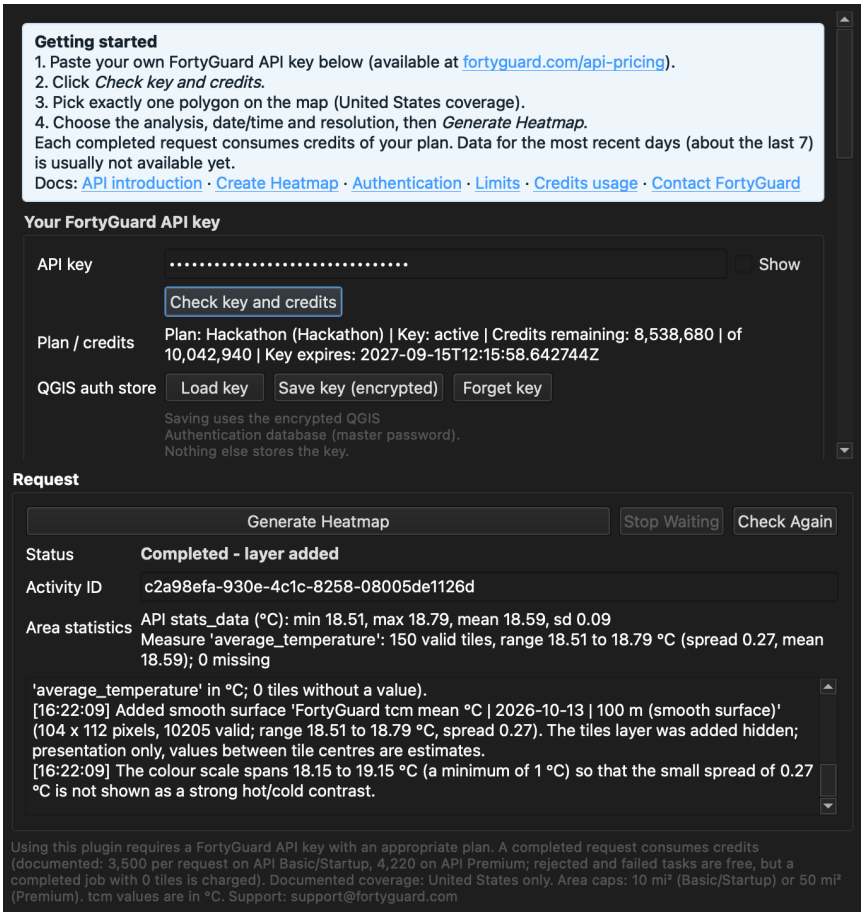


Figure 1. The plugin dialog after a completed production request with the default date (QGIS 3.44.15, macOS): live credits, coverage status in the selected-feature line, activity ID, statistics with the data spread.

6. Review screenshots requested after the first report

The report review asked for a better heatmap: more granular, smooth, more classes, a larger area and time frame, in San Jose. The request below was made through the plugin's own code against the production API: downtown San Jose, a 4 km by 4 km square aligned to the UTM grid (16 km2, about 6.2 mi2), 60 m resolution, 15 to 21 July 2024 (a seven-day window), 4,290 tiles.

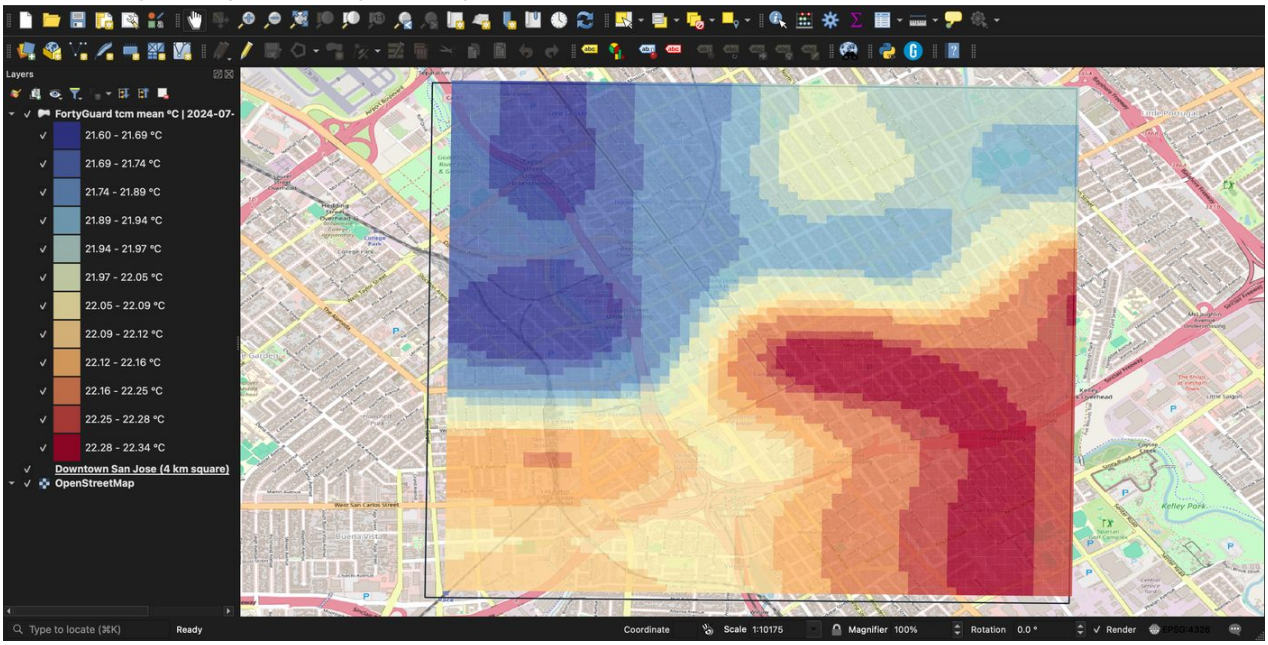


Figure 2. Mean temperature of the week, exact API tiles with 12 legend classes over OpenStreetMap, inside the QGIS window.

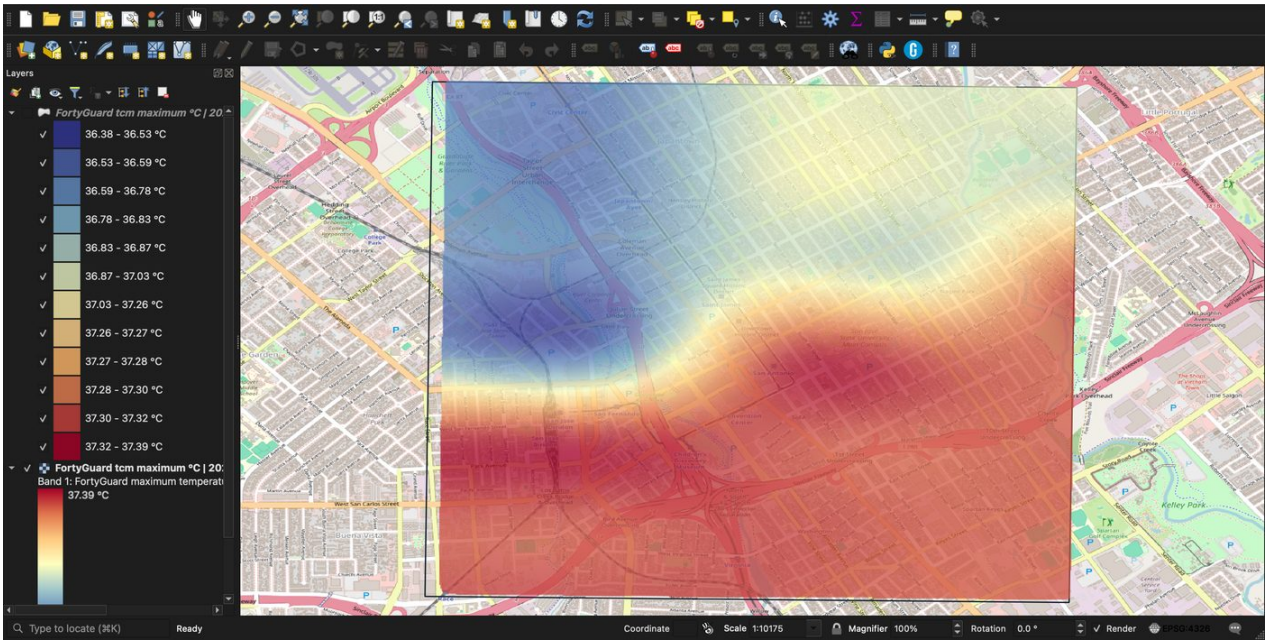


Figure 3. Maximum temperature of the week as the smooth surface (interpolated estimate; the tiles layer is kept underneath). The legend heading names the quantity and unit.

7. Releases, evidence and next steps

Version	Date	Content	Link
0.5.0	20 Oct 2026	Round-one fixes (26 issues); no longer experimental	github.com/FortyGuard-Tech/qgis-fortyguard-heatmap/releases/tag/v0.5.0
0.5.1	20 Oct 2026	Legend classes option (report review)	.../releases/tag/v0.5.1
0.5.2	20 Oct 2026	Round-two fixes (Windows)	.../releases/tag/v0.5.2
0.5.3	20 Oct 2026	Independent code review fixes; version to publish	.../releases/tag/v0.5.3

Evidence in the repository

- docs/TEST_REPORT.md, sections 0 to 0d: issue / root cause / fix / test tables for every row, and the production verification log.
- docs/API_CONTRACT.md, "Observed behaviour": the API facts behind the fixes, for the API team.
- CHANGELOG.md: per-version notes. tests/: 201 automated tests. dist/live_dialog_report.json: the step-by-step record of the production dialog run.
- docs/report/images/: the review screenshots (Phoenix and San Jose, 12 classes, smooth surface, zoom, full-contrast comparison).

Next steps

- Publish 0.5.3 on plugins.qgis.org (Add version) and ask the team to re-test on the published build.
- Fawad to re-confirm the time-zone helper text (inputs are the area's local time; peak hours are UTC).
- API team: confirm which usage field is authoritative for the balance, and whether the 7-day data latency and the flat 4,220-credit charge for empty results are intended.
- QGIS 4 (Qt 6) remains untested.